

# Comparison of Numerical Algorithm for Linear Models

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## 1 Preliminaries

**Definition 1** (Condition Number).

$$\kappa(A) := \|A\|_{op} \|A^{-1}\|_{op}$$

, where  $\|\cdot\|_{op}$  denotes the operator norm.

**Definition 2** (Relative Error).

$$\text{RelErr} = \frac{\|\hat{\beta} - \beta_{\text{true}}\|_2}{\|\beta_{\text{true}}\|_2}$$

## 2 Theoretical Analysis for Condition Numbers

This section compares the numerical stability of solving the Least Squares problem using Normal Equations (Direct Normal Equations and Cholesky Decomposition) and QR Decomposition (Classical Gram-Schmidt and Householder Reflection).

The stability of a linear system  $A\beta = b$  is governed by the condition number  $\kappa(A) = \|A\| \|A^{-1}\|$ . For the Euclidean norm,  $\kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ , where  $\sigma$  denotes singular values.

### 2.1 Normal Equations

The Normal Equations method solves  $(X^T X)\beta = X^T y$ . The computational accuracy depends on the condition number of the matrix  $X^T X$ .

**Lemma 1.** *The condition number of the Normal Equations is the square of the condition number of the design matrix  $X$ .*

*Proof.* Let the Singular Value Decomposition of  $X$  be  $X = U\Sigma V^T$ , where  $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p)$ . Then,

$$X^T X = (V\Sigma U^T)(U\Sigma V^T) = V\Sigma^2 V^T.$$

The singular values of  $X^T X$  are  $\sigma_i^2$ . Consequently,

$$\kappa_2(X^T X) = \frac{\sigma_{\max}^2}{\sigma_{\min}^2} = \left( \frac{\sigma_{\max}}{\sigma_{\min}} \right)^2 = (\kappa_2(X))^2.$$

□

## 2.2 QR Decomposition

The QR method decomposes  $X = QR$ , where  $Q$  is orthogonal and  $R$  is upper triangular. The solution is obtained by solving the triangular system  $R\beta = Q^T y$ .

**Lemma 2.** *The QR method depends on  $\kappa_2(X)$ , not  $\kappa_2(X)^2$ .*

*Proof.* The stability of back-substitution depends on  $\kappa_2(R)$ . Since  $Q$  is orthogonal ( $Q^T Q = I$ ), it preserves the Euclidean norm ( $\|Qv\|_2 = \|v\|_2$ ). Thus, the singular values of  $R$  are identical to those of  $X$ :

$$X = QR \implies \sigma_i(X) = \sigma_i(QR) = \sigma_i(R).$$

Therefore,

$$\kappa_2(R) = \frac{\sigma_{\max}(R)}{\sigma_{\min}(R)} = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)} = \kappa_2(X).$$

□

## 3 Numerical Experiments

We simulated a linear model  $y = X\beta + \epsilon$  with  $n = 200$ ,  $q = 10$ ,  $\beta_{true} = \mathbf{1}$ , and  $\epsilon \sim \mathcal{N}(0, 0.1^2 I)$ . Two design matrices were generated:

- **Well Conditioned Matrix ( $X$ ):**  $X_{ij} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ .
- **Ill-conditioned Matrix ( $X^\natural$ ):** Columns constructed as  $X_{\cdot j}^\natural = z + \delta_j$  (where  $z \sim \mathcal{N}_n(0, I)$  and  $\delta_j \sim \mathcal{N}_n(0, 0.01^2 I_n)$ ).

### 3.1 Condition Numbers

Table 1 highlights the extreme instability of  $X^\natural$ , particularly for Normal Equation methods where  $\kappa$  is squared.

Table 1: Condition Numbers

| Case             | $\kappa(X)$ (QR)   | $\kappa(X^T X)$ (Normal Eq.) |
|------------------|--------------------|------------------------------|
| Well-conditioned | $1.40 \times 10^0$ | $1.92 \times 10^0$           |
| Ill-conditioned  | $3.88 \times 10^2$ | $1.51 \times 10^5$           |

### 3.2 Relative Errors

Table 2 compares the relative errors of each methods.

Table 2: Relative Errors

| Algorithm                          | Well Cond.            | Ill Cond.             |
|------------------------------------|-----------------------|-----------------------|
| 1. Direct Inverse $((X^T X)^{-1})$ | $8.09 \times 10^{-3}$ | $5.78 \times 10^{-1}$ |
| 2. Cholesky $(LL^T)$               | $8.09 \times 10^{-3}$ | $5.78 \times 10^{-1}$ |
| 3. Gram-Schmidt QR                 | $8.09 \times 10^{-3}$ | $5.78 \times 10^{-1}$ |
| 4. Householder QR                  | $8.09 \times 10^{-3}$ | $5.78 \times 10^{-1}$ |

The fact that the results coincide in the ill-conditioned case may seem unexpected. We provide a careful explanation of this phenomenon in the next section.

## 4 Sensitivity Analysis

In the presence of noise  $\epsilon$ , the total error in the least squares solution is dominated by the amplification of noise due to the ill-conditioned nature of  $X$ , rather than the numerical precision of the algorithm. (as mentioned in Section 9.4 of Tibshirani [2015])

Consider the linear model  $y = X\beta_{\text{true}} + \epsilon$ . The least squares estimator  $\hat{\beta}$  is given by:

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T y \\ &= (X^T X)^{-1} X^T (X\beta_{\text{true}} + \epsilon) \\ &= \underbrace{(X^T X)^{-1} X^T X}_{I} \beta_{\text{true}} + \underbrace{(X^T X)^{-1} X^T}_{X^\dagger} \epsilon\end{aligned}$$

Thus, the difference between the estimated and true coefficient is:

$$\hat{\beta} - \beta_{\text{true}} = X^\dagger \epsilon$$

where  $X^\dagger$  denotes the Moore-Penrose pseudoinverse.

Let  $\sigma_{\min}$  be the smallest singular value of  $X$ . And note that  $\|X^\dagger\|_{op} = 1/\sigma_{\min}$ . The norm of the error is bounded by:

$$\|\hat{\beta} - \beta_{\text{true}}\|_2 = \|X^\dagger \epsilon\|_2 \leq \|X^\dagger\|_{op} \|\epsilon\|_2 = \frac{1}{\sigma_{\min}} \|\epsilon\|_2$$

Since  $X$  is ill-conditioned,  $\sigma_{\min} \approx 0$ , causing the factor  $\frac{1}{\sigma_{\min}}$  to be extremely large. If we set  $\epsilon \approx 0$ , then the result is as following.

Table 3: Relative Errors

| Algorithm                          | Ill Cond. ( $\epsilon \sim \mathcal{N}(0, 10^{-16}I)$ ) | Ill Cond. ( $\epsilon \sim \mathcal{N}(0, 0.1^2I)$ ) |
|------------------------------------|---------------------------------------------------------|------------------------------------------------------|
| 1. Direct Inverse $((X^T X)^{-1})$ | $2.83 \times 10^{-11}$                                  | $5.78 \times 10^{-1}$                                |
| 2. Cholesky $(LL^T)$               | $2.52 \times 10^{-11}$                                  | $5.78 \times 10^{-1}$                                |
| 3. Gram-Schmidt QR                 | $1.02 \times 10^{-11}$                                  | $5.78 \times 10^{-1}$                                |
| 4. Householder QR                  | $4.95 \times 10^{-14}$                                  | $5.78 \times 10^{-1}$                                |

## 5 Conclusion

While all methods exhibit comparable accuracy for well-conditioned data, Householder QR proves strictly superior for the ill-conditioned case. Notably, the Direct Inverse method (Normal Equations) demonstrated the highest instability. This empirical result aligns with the theoretical analysis in Section 2; since the stability of the Direct Inverse method depends on the squared condition number  $\kappa(X^T X)$ , whereas QR depends on  $\kappa(X)$ , this degradation in performance is to be expected.

## References

Ryan Tibshirani. Lecture 9: Numerical linear algebra primer. Lecture notes for 10-725/36-725: Convex Optimization, Carnegie Mellon University, 2015. URL <https://www.stat.cmu.edu/~ryantibs/convexopt-S15/scribes/09-num-lin-alg-scribed.pdf>. Spring 2015, February 11. Scribes: Avinash Siravuru, Guofan Wu, Maosheng Liu.